



Effects of supplementation with *Saccharomyces cerevisiae* products on dairy calves: A meta-analysis

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ABSTRACT

Saccharomyces cerevisiae products (SCP) have the potential to promote the growth and development of the gastrointestinal tract and immunity in young livestock animals. However, the effects of SCP supplementation on calves are inconsistent among the reported studies in the literature. Hence, we performed a meta-analysis to comprehensively assess the effects of SCP on the growth performance, ruminal fermentation parameters, nutrients digestibility, ruminal histological morphology, serum immune response, and fecal pathogen colony counts in calves. We searched the Web of Science, ScienceDirect, PubMed, and China National Knowledge Infrastructure for relevant studies published up to October 1, 2021. After screening against a set of criteria, the data of 36 studies were included in our meta-analysis (2,126 calves in total). We evaluated the quality of the data using sensitivity analysis and assessed publication bias. Our meta-analysis revealed several important findings. First, SCP supplementation increased the ruminal short-chain fatty acid concentration, ruminal papilla height, and fiber digestibility, pointing toward stimulation of the development of the rumen in calves. Second, SCP supplementation increased the serum concentrations of total protein, IgA, and IgG but decreased fecal pathogen colony counts, suggesting that SCP could help calves to promote immunity (especially maintaining circulating concentrations of immunoglobulins in preweaning calves) and resistance to pathogens. Third, a subgroup analysis between preweaning and postweaning calves showed that SCP increased average daily gain and dry matter intake preweaning but not postweaning, suggesting that SCP is better supplemented to preweaning calves

to achieve the best results. Fourth, based on the dose-response curve, 24 to 25 g/d might be the optimal dose range of SCP supplementation (into starter feed) preweaning to achieve the best overall effect, meanwhile, we need more studies to improve the consistency and accuracy of the dose-response curve prediction. Overall, SCP supplementation improved growth performance, rumen development, and immunocompetence in calves, particularly in preweaning calves.

Key words: meta-analysis, dairy calves, *Saccharomyces cerevisiae* products, calf health

INTRODUCTION

Proper feeding and management of dairy calves have important effects on the sustainability and profitability of a dairy farm (Soberon et al., 2012). For newborn calves, early feeding regimens (Žitňan et al., 1999) and nutritional interventions (Anderson et al., 1987) have a significant influence on the development of the rumen and the rumen microbiome. The development of both directly affects feed intake, digestibility, and eventually the growth of calves, affecting subsequent growth, health, and milk production performance (Diao et al., 2019). Hence, the initial rumen development is one of the most important focus areas of calf nutrition (Diao et al., 2019). Additionally, although preventive strategies such as proper hygiene may be practiced on farms, infection with pathogens is still inevitable in many intensive-rearing systems (Fuller, 1989). This is mainly due to the undeveloped immune system (Famulener, 1912) and gastrointestinal microbiome of calves (Martin et al., 2021). Antibiotics are widely used to prevent and treat infectious diseases, especially diarrhea, and promote the growth of calves (Brown et al., 2017). However, antibiotics kill not only pathogenic microbes but also commensal and beneficial microbes impairing the establishment of a healthy gut microbiome, which may lead to some diseases (Sullivan et al., 2001). Additionally, the emergence of antibiotic-resistant bacteria and antibiotic residues

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in meat and milk poses a risk to the health of both animals and humans (Stanton, 2013). Hence, nonantibiotic alternatives are sought after in the feeding and management of calves.

Live *Saccharomyces cerevisiae*, its fermentation products, and cell wall have been used as feed additives in animal production (Lynch and Martin, 2002). Previous studies have shown that supplementation with *Saccharomyces cerevisiae* products (SCP; either live yeast or its fermentation products) can enhance the resistance to pathogenic bacteria in vitro (Pérez-Sotelo et al., 2005) and modulate the gut microbiome in dairy calves (Xiao et al., 2016). Hence, SCP have the potential as common feed additives for cattle, pigs, sheep, in both young and adult animals (Kiarie et al., 2011; Zaworski et al., 2014). However, mixed results were reported in previous studies about the effects of SCP supplementation on calves, especially on growth performance (Alugongo et al., 2017a). Indeed, some studies found that SCP supplementation significantly improved ADG and DMI in calves (Hassan et al., 2016; Stefanska et al., 2018), whereas others did not observe any significant benefits (He et al., 2017; Saldana et al., 2019) or even observed decrease in ADG and DMI in calves fed SCP (Vendramini and Arthington, 2007; Mitchell and Heinrichs, 2020). Mitchell and Heinrichs (2020) found that SCP supplementation did not have any effects on feed intake, growth, or nutrient digestibility in weaned calves (aged 7–16 wk) fed different forages. In another study (Fomenky et al., 2017), feeding live yeast at 7.50×10^8 cfu/d to preweaning calves increased ruminal total lactobacilli population and altered colon morphology but did not improve growth performance. Different factors such as nursing stage (Galvão et al., 2005), pathogenic infection (Watanabe et al., 2019), mode of supplementation (starter vs. milk) (Galvão et al., 2005), and calf breeds (Hill et al., 2009) can also affect the efficacy of SCP. The mode of action of SCP is still not entirely clear in calves (Alugongo et al., 2017a), and the benefits of SCP in calves are still debatable. Although the aforementioned factors likely contributed to the variable effects or benefits, the lack of consistent outcomes makes it difficult for animal nutritionists to comprehensively understand and further evaluate the effects of SCP and recommend the use of SCP on calves.

Compared with qualitative literature reviews, meta-analysis as a quantitative alternative can provide objective evidence to resolve mixed results in research and evaluate the effect of treatments (Viechtbauer, 2010). In the present study, we conducted a meta-analysis to evaluate the effect of SCP supplementation on the growth performance, nutrient digestibility, ruminal his-

tological morphology, ruminal fermentation parameters, serum immune response, and fecal pathogen colony counts in calves. Based on the meta-analysis, the nursing stage and optimal amount of SCP supplementation for calves were recommended.

MATERIALS AND METHODS

Literature Search and Selection

We searched PubMed (<https://pubmed.ncbi.nlm.nih.gov>), ScienceDirect (<http://www.sciencedirect.com>), Web of Science (<http://webofknowledge.com>), and China National Knowledge Internet (<http://www.cnki.net>) for relevant studies using a set of key words (Table 1). All relevant articles archived in the 4 databases up to October 1, 2021, were downloaded. According to the participant, intervention, comparison, outcome (PICO) principle (Liberati et al., 2009), we carefully read the retrieved articles to determine whether the studies have the needed data and should be included in the final meta-analysis. In total, 36 studies were included (Supplemental Table S1; <https://doi.org/10.6084/m9.figshare.20166629.v1>; Zhang, 2022). Our literature retrieval process and PICO principle are shown in Supplemental Figure S1 (<https://doi.org/10.6084/m9.figshare.20166629.v1>; Zhang, 2022). Calves receiving no SCP supplementation were considered as the control group. Two researchers independently evaluated the studies to be included, and disagreements were resolved by a third investigator.

Information Extraction

The information of the included studies including the publications (the first author, year of publication, and publishing journal), calves (breed, sex, age, initial BW, weaning age, nursing stage, and sample size), and experimental design [types of SCP used (yeast fermentation products or live yeast), method of addition (added to feedstuff, milk, or as capsules), dose, duration of SCP feeding, and experimental outcomes] were recorded. The above information was summarized in Supplemental Table S1.

According to the general experimental outcomes of the included studies, we included the following data in this meta-analysis: growth performance (ADG and DMI), ruminal fermentation parameters [concentrations short-chain fatty acids (SCFA) and $\text{NH}_3\text{-N}$ concentration], nutrient digestibility [DM digestibility (DMD), CP digestibility (CPD), NDF digestibility (NDFD), and ADF digestibility (ADFD)], ruminal histological morphology (ruminal papilla height and weight), the

Table 1. Literature search strategy for relevant studies

Search	Query	Items found
PubMed (https://pubmed.ncbi.nlm.nih.gov)		
1	(<i>Saccharomyces</i> [title/abstract]) OR (yeast [title/abstract]) OR (Baker's yeast [title/abstract]) OR (<i>S. cerevisiae</i> [title/abstract]) OR (Brewer's yeast [title/abstract])	212,688
2	(calf [Title/Abstract]) OR (calves [Title/Abstract])	65,655
1 AND 2		441
Web of Science (http://webofknowledge.com)		
1	TS = (<i>Saccharomyces</i> OR yeast OR yeast OR <i>S. cerevisiae</i> OR Brewer's Yeast)	424,985
2	TS = (calf OR calves)	136,564
1 AND 2		1,668
ScienceDirect (http://www.sciencedirect.com)		
1	(<i>Saccharomyces</i> OR yeast OR <i>S. cerevisiae</i> OR Brewer's yeast) AND (calf OR calves)	3,264
CNKI (http://www.cnki.net)		
1	"calf" AND "yeast"	305

titer of serum antibodies (IgA and IgG), serum total protein (TP), and counts of fecal pathogens (*Salmonella*, *Clostridium*, and *Escherichia coli*). For the meta-analysis, the counts of fecal pathogens (cfu/g) were expressed as $\log_{10}$ cfu/g to meet the requirement for a normal distribution of residuals for statistical analysis. The descriptive statistics (n, mean, median, maximum, and minimum values) for the outcomes were shown in Table 2. The mean values of these data, standard deviation (SD), and the standard error of means between groups were recorded. We used pooled SD, which was calculated by multiplying the standard error of means by the square root of the number of trials (Xu et al., 2020b), as the within-group SD.

Statistical Analysis

We performed the meta-analysis of the included data using the METAN module of StataSE 14 (Stata Statistical Software, Release 12. StataCorp LP). The Chi-squared (Q) test and the I^2 statistic were used to measure heterogeneity (Higgins and Thompson, 2002). The I^2 statistic describes the percentage of total variation across studies due to heterogeneity rather than chance; it was calculated as follows:

$$I^2 = \left[\frac{Q - (k - 1)}{Q} \right] \times 100,$$

Table 2. Descriptive statistics for outcomes of 36 articles included in the meta-analysis

Outcome ¹	Treatment					Control				
	n	Mean	Median	Maximum	Minimum	n	Mean	Median	Maximum	Minimum
ADG, g/d	1,408	0.76	0.72	1.80	0.18	1,426	0.73	0.74	1.84	0.17
DMI, kg/d	1,031	2.32	1.54	8.40	0.14	1,041	2.31	1.36	8.60	0.15
TSCFA, mM	130	105.30	106.60	186.00	36.00	130	96.80	90.00	152.40	32.30
Acetate, mM	130	48.11	51.38	95.98	19.00	130	43.98	47.26	82.45	17.50
Propionate, mM	130	33.73	33.53	75.23	11.50	130	30.66	23.87	67.16	10.10
Butyrate, mM	130	10.01	8.49	32.18	2.70	130	8.05	6.60	20.39	2.49
NH ₃ -N	74	10.50	9.99	16.90	4.93	74	11.05	11.73	15.70	4.85
DMD, %	102	75.00	76.00	87.00	58.00	102	70.00	73.00	79.00	54.00
CPD, %	78	69.00	67.00	85.00	63.00	78	62.00	60.00	82.00	58.00
NDFD, %	102	59.00	56.00	77.00	41.00	102	52.00	55.00	66.00	32.00
ADFD, %	82	53.00	59.00	62.00	40.00	82	49.00	54.00	56.00	34.00
Papilla height, μ m	158	1,175.30	1,181.00	173.00	236.00	158	1,172.10	1,002.30	2,470.00	202.50
Papilla width, μ m	158	488.50	392.00	920.00	207.60	158	497.90	390.00	880.00	190.70
TP, g/dL	233	5.70	5.65	7.59	5.06	233	5.53	5.52	6.49	5.14
Serum IgA, mg/mL	108	0.42	0.22	1.06	0.19	108	0.41	0.19	0.99	0.18
Serum IgG, mg/mL	108	20.22	16.97	37.70	11.18	108	19.23	17.26	31.34	10.95
Pathogens	167	5.07	0.12	16.20	0.00	167	7.58	5.71	16.90	1.05

¹TSCFA = ruminal total short-chain fatty acids; acetate = ruminal acetate; propionate = ruminal propionate; butyrate = ruminal butyrate; DMD = DM digestibility; CPD = CP digestibility; NDFD = NDF digestibility; ADFD = ADF digestibility; papilla height = ruminal papilla height; papilla width = ruminal papilla width; TP = serum total protein; pathogens = fecal pathogen colony counts [$\log_{10}$ (cfu/g)].

where Q is the χ^2 heterogeneity statistic and k is the number of studies. Significant heterogeneity was declared at $I^2 > 50\%$ or $P_{\text{heterogeneity}} < 0.10$ (Deeks et al., 2008).

The effect sizes included in our study were all continuous variables, hence the standardized mean difference (SMD) was selected as the effect magnitude as follows:

$$\text{SMD} = \frac{\mu_1 - \mu_2}{SP}; SP = \sqrt{\frac{(n_1 - 1)SD_1^2 + (n_2 - 1)SD_2^2}{n_1 + n_2 - 2}},$$

where μ_1 = the mean of SCP supplementation group, μ_2 = the mean of control group, SP = combined SD, SD_1 = the SD of SCP supplementation group, SD_2 = the SD of control group, n_1 = the sample size of SCP supplementation group, and n_2 = the sample size of control group (Deeks et al., 2008).

Due to the heterogeneity ($I^2 > 50\%$) for the effect sizes included in our study, the random-effects model was selected as the pooling model. The 95% confidence interval (CI) was calculated, and the inverse-variance approach calculates a weighted average as

$$\text{generic inverse - variance weighted average} = \frac{\sum Y_i (1/SE_i^2)}{\sum (1/SE_i^2)},$$

where Y_i = the intervention effect estimated in the i study, SE_i = the standard error of that estimate, and the summation is across all studies (Deeks et al., 2008).

To establish a comprehensive and systematic additive assessment system for calves, we followed the following principles: (1) we included as many studies and outcomes as possible; (2) we excluded individual data that had large deviations from the overall mean value using sensitivity analysis (the METANINF module; Supplemental Figures S2–S7; <https://doi.org/10.6084/m9.figshare.20166629.v1>; Zhang, 2022); (3) when 10 or more observations were available reporting the effects on the growth performance, rumen fermentation parameters, nutrient digestibility, serum metabolites, and fecal pathogen colony counts, we used the Egger's test (the METABIAS module) and funnel plot (the METAFUNNEL module) to evaluate publication bias (Egger et al., 1997); and (4) when 15 or more observations were available reporting the effects on the above parameters, we explored the source of heterogeneity using subgroup analyses (Xu et al., 2020a). The subgroups were defined as nursing stage and types of SCP. The figures of meta-analysis were also performed by StataSE 14.

We performed regression analysis to determine the optimal dose range of supplementation of live yeast or its fermentation products. According to the results of subgroup analyses, we used the data of the preweaning stage to perform regression analysis. To uniform the unit supplementation dose, we converted different expressions of supplementation doses to g/d for yeast fermentation products and cfu/d for live yeast, based on DMI. Therein, the studies by Galvão et al. (2005) and He et al. (2007) were excluded where SCP was supplemented into milk replacer and had no effect on calves. The study by Seymour et al. (1995) was also excluded as no dose information was reported. Finally, the data in the dose-response equations were only for SCP that were added to solid feed.

Unary linear regression was chosen as the model to explore the relationship between the dose of SCP (yeast fermentation products or live yeast) supplementation and growth performance (ADG or DMI) in calves. Because the data used in the regression equations were collected from multiple studies, the random effects of individual studies needed to be accounted for in the modeling process. Therefore, random effects were incorporated into the estimation procedure using a mixed model framework. The general form of the mixed model is as follows (St-Pierre, 2001):

$$Y_{ij} = B_0 + B_1X_{ij} + B_2X_{ij}^2 + s_i + b_iX_{ij} + e_{ij},$$

where Y_{ij} = dependent variable of weighted SMD of ADG or DMI, B_0 = overall intercept across all studies, B_1 = linear regression coefficient of Y on X (fixed effect), B_2 = quadratic regression coefficient of Y on X (fixed effect), X_{ij} = value of the predictor variable (dose of yeast fermentation products or live yeast), s_i = random effect of study i , b_i = random effect of study i on the regression coefficient of Y on X in study i , and e_{ij} = the unexplained residual error. The linear or quadratic model that had lower P -values and higher R^2 was used in the following analysis.

To account for variations in precision among the studies, we used the SMD from the meta-analysis as the dependent variable, which is a weighting step. Weighting in this manner can improve the overall precision of the regression estimates and ensure homogeneity of variance for the model (St-Pierre, 2001). The lme4 package in R (Version 4.1.2, <https://cran.r-project.org/web/packages/lme4/index.html>) was used for the mixed model analysis. The ggplot2 package was used to create the figures of regression equations. The probability levels were set at $P < 0.05$ for significance and $0.05 \leq P < 0.10$ for a trend.

RESULTS

Sensitivity Analysis and Publication Bias

The sensitivity analysis showed that all the estimated values were within the CI after excluding a single datum (Supplemental Figures S2–S7). These results showed that the data were robust, and a single study had no effect on the results of the meta-analysis. If 10 or more trials were available, we examined their publication bias by using funnel plots (Figure 1). Except for IgG ($P_{Egger} = 0.60$), publication bias was noted for all the outcomes ($P_{Egger} \leq 0.03$; Figure 1). Hence, the random-effects model was selected as the pooling model.

Effect of SCP on Growth Performance and Nutrient Digestibility

Increase in DMI ($P < 0.001$) and ADG ($P = 0.013$) was evident in calves fed SCP (Table 3). However, these responses had high significant heterogeneity. Supplementation with SCP increased DMD ($P < 0.001$), CPD ($P < 0.001$), NDFD ($P < 0.001$), and ADFD ($P = 0.004$). The above effects also displayed significant heterogeneity ($I^2 > 50.00\%$, $P_{heterogeneity} < 0.10$) among the included studies (Table 3). The numbers of observations on DMD ($n = 11$), CPD ($n = 8$), NDFD ($n = 11$), and ADFD ($n = 9$) were less than 15, so we did not perform subgroup analyses on these outcomes.

Subgroup analyses (nursing stage) showed that SCP supplementation improved ADG ($P = 0.027$) and DMI ($P < 0.001$) among preweaning calves, but not among postweaning calves ($P = 0.363$ for ADG, $P = 0.483$ for DMI) (Table 4). No significant heterogeneity with respect to the effect on ADG ($I^2 = 41.80\%$, $P_{heterogeneity} = 0.12$) or DMI ($I^2 = 28.20\%$, $P_{heterogeneity} = 0.11$) was noted among the included studies in the postweaning period (Table 4).

Subgroup analyses (types of SCP) showed that live yeast supplementation could improve DMI ($P < 0.001$) but only an increasing trend for ADG ($P = 0.056$). Yeast fermentation product supplementation (with feedstuff, milk, or capsule) had no effect on ADG ($P = 0.161$) or DMI ($P = 0.152$). The subgroup analyses revealed significant heterogeneity in the effect on ADG and DMI ($P_{heterogeneity} < 0.10$; Table 5).

Effect of SCP on Ruminal Fermentation Parameters and Histological Morphology

Supplementation with SCP increased the concentration of total SCFA ($P = 0.010$), acetate ($P = 0.031$), propionate ($P = 0.001$), and butyrate ($P = 0.043$) in the rumen. No significant effect on $\text{NH}_3\text{-N}$ concentra-

tion was noted ($P = 0.148$). There was significant heterogeneity ($I^2 > 50.00\%$, $P_{heterogeneity} < 0.10$) in the rumen fermentation characteristics among the studies (Table 3).

Supplementation with SCP increased the height of ruminal papilla ($P = 0.029$), but no significant effect on the width of ruminal papillae ($P = 0.684$) was noticed. Significant heterogeneity ($P_{heterogeneity} < 0.10$) in the effect on these 2 measurements was revealed among the studies (Table 3). The numbers of observations of height ($n = 9$) and width of ruminal papillae ($n = 9$) were less than 15, so we did not perform subgroup analyses.

Similar to the growth performance data, SCP supplementation increased the concentration of ruminal total SCFA, and acetate, propionate, and butyrate preweaning ($P \leq 0.011$), but not postweaning ($P \geq 0.140$; Table 4). Supplementation with live yeast ($P \leq 0.018$), but not yeast fermentation products ($P \geq 0.140$), increased the concentrations of total SCFA, acetate, and propionate (Table 5).

Effect of SCP on Serum Metabolites and Fecal Pathogen Colony Counts

Compared with the control, SCP supplementation increased serum content of TP (5.53 vs. 5.70 g/dL, $P = 0.003$), IgG (19.23 vs. 20.22 mg/mL, $P = 0.011$), and IgA (0.41 vs. 0.42 mg/mL, $P = 0.001$) in the calves. The effect on the content of TP ($I^2 = 74.90\%$, $P_{heterogeneity} < 0.001$) and IgA ($I^2 = 89.70\%$, $P_{heterogeneity} < 0.01$) exhibited significant heterogeneity, but not on that of IgG ($I^2 = 34.20\%$, $P_{heterogeneity} = 0.13$).

Supplementation with SCP decreased fecal pathogens ($P < 0.001$). However, the included studies showed significant heterogeneity ($I^2 = 94.10\%$, $P_{heterogeneity} < 0.01$) in the magnitude of pathogen decrease among the studies (Table 3). Interestingly, compared with other pathogens, *E. coli* was decreased to a much greater extent than other pathogens by SCP supplementation (Supplemental Figure S8; <https://doi.org/10.6084/m9.figshare.20166629.v1>; Zhang, 2022).

Subgroup analyses showed that supplementation with SCP increased TP in preweaning calves ($P = 0.003$), but not postweaning calves ($P = 0.358$; Table 4). Supplementation with live yeast ($P = 0.003$), but not yeast fermentation products ($P = 0.358$), increased serum TP (Table 5).

Dose-Response Relationship Between SCP and Growth Performance

According to the results of subgroup analysis (nursing stage), the effect of SCP supplementation was only

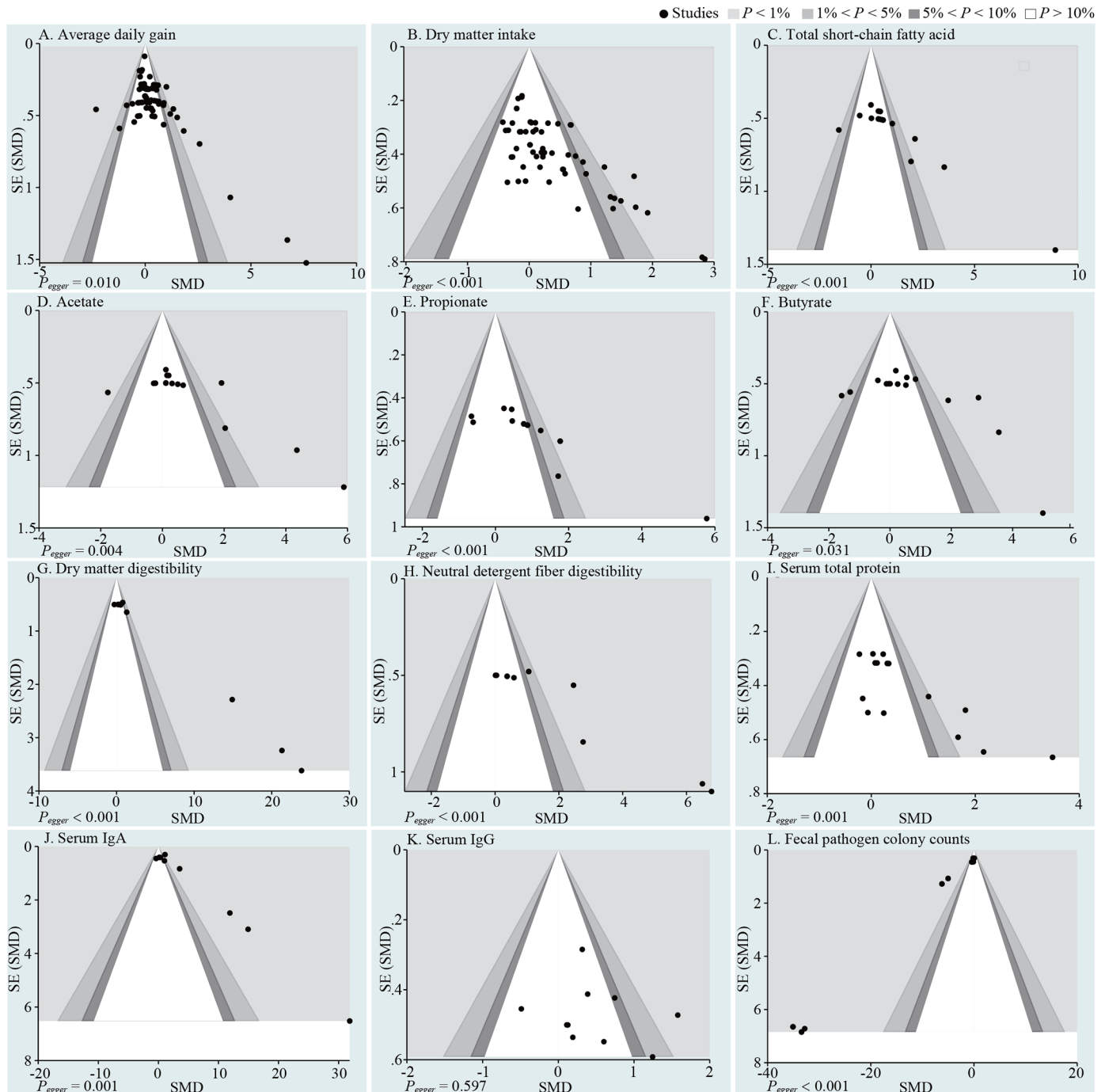


Figure 1. The publication bias about the effect of supplementation with *Saccharomyces cerevisiae* products on calves. The number of controlled trials reporting ADG (A), DMI (B), total short-chain fatty acids (C), acetate (D), propionate (E), butyrate (F), DM digestibility (G), NDF digestibility (H), serum total protein (I), serum IgA (J), serum IgG (K), and fecal pathogen colony counts (L) was each more than 10. We used the funnel plots and Egger's test to determine publication bias. The asymmetry funnel plots and the $P_{egger} < 0.05$ indicate publication bias. SMD = standardized mean difference, SE (SMD) = standard error (standardized mean difference). For the analysis process, the counts of fecal pathogens (cfu/g) were expressed as Log_{10} cfu/g.

significant in preweaning calves (Table 4), and hence the dose-response equations were only applicable to preweaning calves. Live yeast and yeast fermentation products, both added to solid feed, exhibited different

dose-response relationships in preweaning calves (Figure 2; Table 6). Yeast fermentation products exhibited a quadratic effect on ADG [$R^2(\text{marginal}) = 0.682$, $\hat{\sigma}_e = 0.564$, $P(\text{Linear}) < 0.001$, $P(\text{Quadratic}) = 0.001$] and

Table 3. Effect size of *Saccharomyces cerevisiae* product supplementation in calves

Item ¹	Numbers of observations	Numbers of studies	Effect size		Heterogeneity	
			SMD (95% CI) ²	P-value	I ² /%	P-value
ADG, g/d	69	34	0.18 (0.04, 0.33)	0.013	65.0	<0.001
DMI, kg/d	59	28	0.27 (0.13, 0.41)	<0.001	54.5	<0.001
TSCFA, mM	15	8	0.88 (0.21, 1.54)	0.010	82.4	<0.001
Acetate, mM	15	8	0.68 (0.06, 1.29)	0.031	80.1	<0.001
Propionate, mM	15	8	0.88 (0.35, 1.42)	0.001	74.3	<0.001
Butyrate, mM	15	8	0.61 (0.03, 1.25)	0.043	81.5	<0.001
NH ₃ -N	8	7	-0.70 (-1.64, 0.25)	0.148	84.7	<0.001
DMD, %	11	5	2.62 (1.24, 4.00)	<0.001	92.2	<0.001
CPD, %	8	4	5.52 (3.13, 7.91)	<0.001	94.7	<0.001
NDFD, %	11	5	1.74 (0.79, 2.69)	<0.001	87.0	<0.001
ADFD, %	9	4	1.12 (0.35, 1.88)	0.004	78.7	<0.001
Papilla height, µm	9	8	0.66 (0.07, 1.26)	0.029	83.4	<0.001
Papilla width, µm	9	8	0.07 (-0.26, 0.40)	0.684	49.7	0.044
TP, g/dL	15	7	0.60 (0.20, 0.99)	0.003	74.9	<0.001
Serum IgA, mg/mL	10	5	1.91 (0.74, 3.08)	0.001	89.7	<0.001
Serum IgG, mg/mL	10	5	0.45 (0.10, 0.80)	0.011	34.2	0.134
Pathogens	13	5	-5.41 (-7.29, -3.53)	<0.001	94.1	<0.001

¹TSCFA = ruminal total short-chain fatty acids; acetate = ruminal acetate, propionate = ruminal propionate, butyrate = ruminal butyrate; DMD = DM digestibility, CPD = CP digestibility (%), NDFD = NDF digestibility; ADFD = ADF digestibility; papilla height = ruminal papilla height; papilla width = ruminal papilla width; TP = serum total protein; pathogens = fecal pathogen colony counts [log₁₀ (cfu/g)].

²SMD = standardized mean difference. The values in parentheses are 95% lower and 95% upper CI.

DMI [$R^2(\text{marginal}) = 0.330$, $\hat{\sigma}_e = 0.819$, $P(\text{Linear}) = 0.025$, $P(\text{Quadratic}) = 0.034$] preweaning, with the effect on ADG peaking at around 25.37 g/d and the effect on DMI peaking at around 24.54 g/d. There was no linear effect on ADG ($P = 0.070$) or quadratic effect on DMI ($P = 0.375$); P -values (*Intercepts*) for the intercepts of any equation were higher than 0.10.

DISCUSSION

In this study, we used a meta-analysis to evaluate the effects of SCP on the growth performance, ruminal fermentation parameters, rumen histological morphology, nutrients digestibility, serum metabolites, and fecal pathogen colony counts in calves. For

Table 4. Subgroup analyses (preweaning and postweaning) of effects of yeast *Saccharomyces cerevisiae* products on calves

Subgroup ¹	Item ²	Numbers of observations	Numbers of studies	Effect size ³		Heterogeneity test ⁵	
				SMD (95% CI) ⁴	P-value	I ² /%	P-value
Preweaning	ADG, g/d	38	24	0.12 (0.01, 0.22)	0.027	73.4	<0.001
	DMI, kg/d	34	20	0.43 (0.22, 0.64)	<0.001	59.8	<0.001
	TSCFA, mM	7	4	1.91 (0.64, 3.17)	0.003	87.9	<0.001
	Acetate, mM	7	4	1.49 (0.35, 2.63)	0.011	86.3	<0.001
	Propionate, mM	7	4	1.45 (0.49, 2.41)	0.003	81.6	<0.001
	Butyrate, mM	7	4	0.70 (0.29, 1.12)	0.001	89.5	<0.001
	TP, g/dL	11	6	0.81 (0.27, 1.34)	0.003	81.4	<0.001
Postweaning	ADG, g/d	28	15	0.06 (-0.06, 0.17)	0.363	41.8	0.120
	DMI, kg/d	22	11	0.06 (-0.10, 0.22)	0.483	28.2	0.108
	TSCFA, mM	7	4	0.19 (-0.53, 0.91)	0.607	68.7	0.004
	Acetate, mM	7	4	0.11 (-0.62, 0.84)	0.770	69.3	0.003
	Propionate, mM	7	4	0.49 (-0.16, 1.14)	0.140	61.3	0.017
	Butyrate, mM	7	4	0.13 (-0.27, 0.52)	0.527	58.2	0.026
	TP, g/dL	4	2	0.17 (-0.20, 0.55)	0.358	0.0	0.927

¹The number of trials for the subgroup analyses was more than 15 (Table 3) for each nursing stage.

²TSCFA = ruminal total short-chain fatty acids; acetate = ruminal acetate; propionate = ruminal propionate; butyrate = ruminal butyrate; TP = serum total protein.

³Effect size was calculated using a random-effects model.

⁴The values in parentheses are 95% lower and 95% upper CI. SMD = standardized mean difference.

⁵The heterogeneity between subgroups: $P_{\text{heterogeneity}}$ of ADG = 0.248, $P_{\text{heterogeneity}}$ of DMI = 0.002, $P_{\text{heterogeneity}}$ of TSCFA = 0.021, $P_{\text{heterogeneity}}$ of acetate = 0.046, $P_{\text{heterogeneity}}$ of propionate = 0.103, $P_{\text{heterogeneity}}$ of butyrate = 0.048, $P_{\text{heterogeneity}}$ of TP = 0.057.

Table 5. Subgroup analyses (live yeast and yeast fermentation products) of *Saccharomyces cerevisiae* supplementation in calves

Subgroup ¹	Item ²	Numbers of observations	Numbers of studies	Effect size ³		Heterogeneity test ⁵	
				SMD (95% CI) ⁴	P-value	I ² /%	P-value
Live yeast	ADG, g/d	22	13	0.38 (−0.01, 0.77)	0.056	80.9	<0.001
	DMI, kg/d	22	12	0.53 (0.24, 0.81)	<0.001	60.6	<0.001
	TSCFA, mM	5	4	1.59 (0.53, 2.65)	0.003	72.3	0.006
	Acetate, mM	5	4	2.28 (0.40, 4.17)	0.018	88.9	<0.001
	Propionate, mM	5	4	1.26 (0.75, 1.77)	0.003	0.0	0.474
	Butyrate, mM	5	4	1.33 (−0.91, 3.58)	0.245	92.3	<0.001
	TP, g/dL	7	4	1.36 (0.46, 2.26)	0.003	84.7	<0.001
Yeast fermentation products	ADG, g/d	43	21	0.09 (−0.04, 0.22)	0.161	41.7	0.003
	DMI, kg/d	36	16	0.10 (−0.04, 0.25)	0.152	42.4	0.004
	TSCFA, mM	9	4	0.62 (−0.29, 1.53)	0.181	85.0	<0.001
	Acetate, mM	9	4	0.18 (−0.42, 0.77)	0.555	70.1	0.001
	Propionate, mM	9	4	0.78 (−0.03, 1.58)	0.140	81.7	<0.001
	Butyrate, mM	9	4	0.46 (−0.10, 1.02)	0.106	66.3	0.003
	TP, g/dL	8	3	0.12 (−0.11, 0.35)	0.358	0.0	0.913

¹The subgroup analyses for numbers of controlled trials was more than 15 (Table 3), according to the type of *Saccharomyces cerevisiae* products.

²TSCFA = ruminal total short-chain fatty acids; acetate = ruminal acetate; propionate = ruminal propionate; butyrate = ruminal butyrate; TP = serum total protein.

³Effect size was calculated using a random-effects model.

⁴The values in parentheses are 95% lower and 95% upper CI. SMD = standardized mean difference.

⁵The heterogeneity between subgroups: $P_{\text{heterogeneity}}$ of ADG = 0.125, $P_{\text{heterogeneity}}$ of DMI = 0.009, $P_{\text{heterogeneity}}$ of TSCFA = 0.172, $P_{\text{heterogeneity}}$ of acetate = 0.037, $P_{\text{heterogeneity}}$ of propionate = 0.322, $P_{\text{heterogeneity}}$ of butyrate = 0.461, $P_{\text{heterogeneity}}$ of TP = 0.009.

a meta-analysis, it is not enough to show the results of effect size only. To establish a comprehensive and systematic assessment, we used sensitivity analysis and Egger's test in the study. Heterogeneity and publication bias were noted among the included studies; hence we used the random-effects model to perform the meta-analysis.

We found considerable heterogeneity in the responses to SCP supplementation among the included studies, and the heterogeneity decreased in the subgroup analysis by splitting postweaning and preweaning effects. However, significant heterogeneity still existed when yeast fermentation products and live yeast were used as subgroups to analyze the resource of heterogeneity. Moreover, we also found that yeast fermentation products and live yeast could have a different influence on ADG and DMI in calves. Although an in vitro study showed that there was no difference in the effect on ruminal fermentation between yeast fermentation products and live yeast (Lynch and Martin, 2002), none of the included studies directly compared the 2 types of products. Hence, we suggest that future studies should consider the differences between the 2 types of products on calves. We further conducted subgroup analyses to explain the heterogeneity in the responses and identified the better method to administer SCP supplementation. Finally, we found that the weaning stage might be the main reason affecting the effects of SCP supplementation on calves. Hence, we mainly focused on the subgroups of preweaning and postweaning.

The whole rumen, especially the rumen wall, plays an important role in SCFA absorption (Penner et al., 2009), regulation of the protons to prevent severe cytosolic acidosis (Gäbel et al., 2002), and ammonia absorption and urea recycling (Abdoun et al., 2006). In our meta-analysis, SCP supplementation increased the concentration of ruminal SCFA. It is generally accepted that ruminal papillary development was profoundly dependent on stimulation by SCFA (Harrison et al., 1960; Newbold and Ramos-Morales, 2020), especially propionate, which is the major substrate for gluconeogenesis in ruminants (Hristov et al., 2013), and butyrate, which is the main energy source of rumen epithelium and plays an important role for rumen epithelial development (Tamate et al., 1962). Our results also revealed that SCP supplementation can increase the height of ruminal papillae, which improves nutrient absorption, ruminal development, nutrient digestibility, and eventually the growth of calves. It is noteworthy that the SCFA are the fermentation products of the rumen microbiome. Hence, the establishment of the rumen microbiome is important for ruminal development, which can increase the digestion of nutrients, especially forage in calves. Our meta-analysis also revealed that SCP supplementation could increase DMD, CPD, NDFD, and ADFD. Based on our results, it can be inferred that SCP supplementation contributes to the establishment of the rumen microbiome. Studies have shown that probiotic supplementation could facilitate the development of the rumen microbiome, which helped

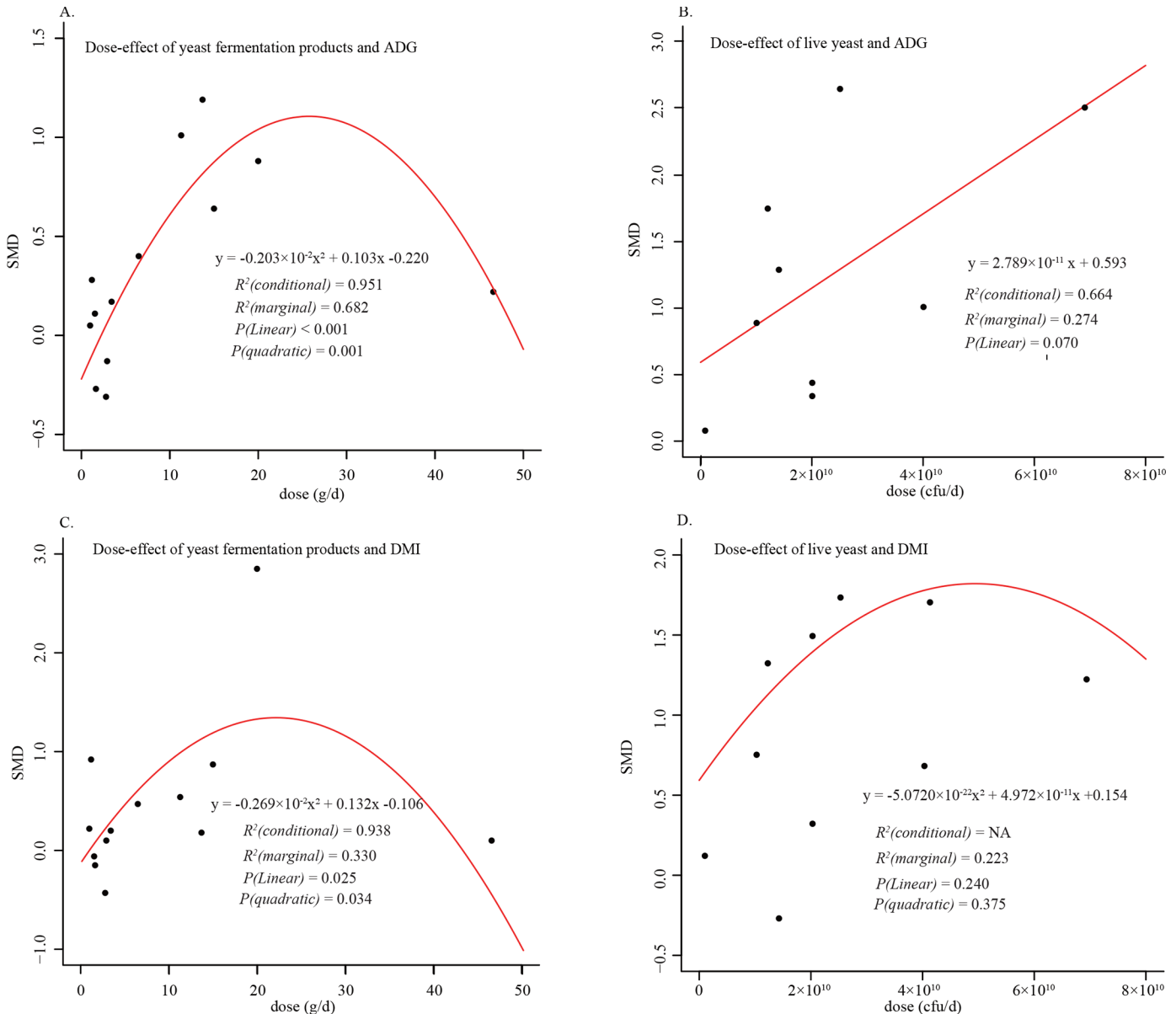


Figure 2. Dose-response relationship between supplementation with *Saccharomyces cerevisiae* products and ADG and DMI. (A) The dose-response relationship between yeast fermentation products and ADG; (B) the dose-response relationship between live yeast and ADG. (C) Dose-response relationship between yeast fermentation products and DMI. (D) Dose-response relationship between live yeast and DMI. SMD = standardized mean difference. NA = missing value.

calves around weaning (Hong et al., 2005). The effects of SCP supplementation on the rumen microbiome may be explained for 3 reasons. First, live *Saccharomyces cerevisiae* can stimulate the growth of total ruminal anaerobes and cellulolytic bacteria by removing oxygen, as demonstrated in vitro (Wallace and Newbold, 1992; Mao et al., 2013). Second, SCP supplementation can increase the abundance of ruminal bacteria involved in lactate metabolism (e.g., *Selenomonas ruminantium* and *Megasphaera elsdenii*), which can further reduce

ruminal lactate, thereby stabilizing ruminal pH and survivability of beneficial bacteria, especially cellulolytic bacteria (Mohammed et al., 2017). Third, yeast cell wall polysaccharides can be used as the substrates for the growth of *Lactobacillus* and *Bifidobacterium* (Karaman et al., 2005) and bind to pathogenic bacteria to reduce their colonization (Kogan and Kocher, 2007), which improves the rumen microbiome. Thus, SCP added to the starter feed may improve the development of both the rumen and the rumen microbiome. Indeed,

Table 6. Parameters of the linear and quadratic models for the dose-response relationship between supplementation with *Saccharomyces cerevisiae* products and ADG and DMI

Equation ¹	Intercept			X			X ²		
	Model ²	Coefficient	SEM	P-value	Coefficient	SEM	P-value	Coefficient	SEM
YFP-ADG	L	0.139	0.019	0.476	0.0139	0.011	0.240	—	—
	Q	-0.220	0.135	0.139	0.103	0.020	<0.001	-0.203 × 10 ⁻²	0.001
YFP-DMI	L	0.328	0.373	0.403	0.0166	0.021	0.246	—	—
	Q	-0.106	0.034	0.765	0.132	0.049	0.025	-0.269 × 10 ⁻²	0.001
LY-ADG	L	0.593	0.449	0.224	2.789 × 10 ⁻¹¹	1.29 × 10 ⁻¹¹	0.070	—	—
	Q	-0.730	0.510	0.195	1.367 × 10 ⁻¹⁰	2.29 × 10 ⁻¹¹	0.250	-1.403 × 10 ⁻²¹	2.80 × 10 ⁻²²
LY-DMI	L	0.539	0.352	0.165	1.459 × 10 ⁻¹¹	1.12 × 10 ⁻¹¹	0.229	—	—
	Q	0.154	0.054	0.783	4.972 × 10 ⁻¹¹	3.87 × 10 ⁻¹¹	0.240	-5.027 × 10 ⁻²²	5.30 × 10 ⁻²²
									σ^2_e
									0.933
									0.564
									0.976
									0.819
									0.726
									0.728
									0.917
									0.881

¹YFP = yeast fermentation products; LY = live yeast.²L = linear model, Q = quadratic model.³ σ^2_e = square root of the estimated residual variance.

one study found that yeast fermentation products added to the milk passed directly to the abomasum, resulting in little or no ruminal effects (Hill et al., 2009). Meanwhile, we suggest that future studies should focus on which components of SCP (live yeast, cell wall, or certain fermentation products or metabolites) are responsible for the observed positive effects on the rumen microbiome of calves.

The immune system of newborn calves is not fully developed, and calves obtain passive immunity through the consumption of colostrum (Franklin et al., 2003). Colostrum primarily contain immune globulins (IgA, IgG, and IgM) that can help clear pathogens (Scott, 2004). In addition to reflecting the body's protein metabolism, serum TP content also serves as an important indicator of the passive immunity status in calves (Aluogongo et al., 2017b). Our meta-analysis showed that SCP supplementation increased serum IgG, IgA, and TP. The increase in IgA is consistent with the finding of Villot et al. (2020) who recently reported that SCP supplementation increased the amount of secretory IgA in the colon and ileum tissues. Immunoglobulin A secretion can affect the intestinal microbiome, help maintain the integrity of the intestinal epithelial barrier, and reduce colonization by pathogens (Pabst et al., 2016). Hence, the positive effect of SCP supplementation may be partially attributed to the enhancement of immunity and thus health in calves. This premise is corroborated by the decreased fecal pathogen colony counts, especially of *E. coli*, upon SCP supplementation. Meanwhile, SCP supplementation may reduce the consumption of intestinal IgG by clearing pathogens, thereby maintaining the circulating concentrations of immunoglobulins. At present, few studies have examined the effect of SCP on the intestines. Future studies are warranted to evaluate how SCP supplementation may affect the small intestines, hindgut, and intestinal immune factors (e.g., sIgA), not just the rumen.

In this study, we found that preweaning calves benefit more from SCP supplementation than weaned calves, especially with respect to ADG and DMI. The positive effect on growth performance can alleviate weaning stress (Di Francia et al., 2008). The reason could be attributed to the positive effect of SCP supplementation on rumen development (both the rumen itself and its microbiome). Additionally, the enhancement of immunity by SCP supplementation shall also help calves to cope with weaning stress (Figure 3). It should be noted that the included studies varied in the effect of SCP supplementation on the growth of calves, as shown by the data included in this meta-analysis. One study found that a high dose (2% of DM) of yeast fermentation products enhanced growth performance and slightly improved rumen development in dairy calves

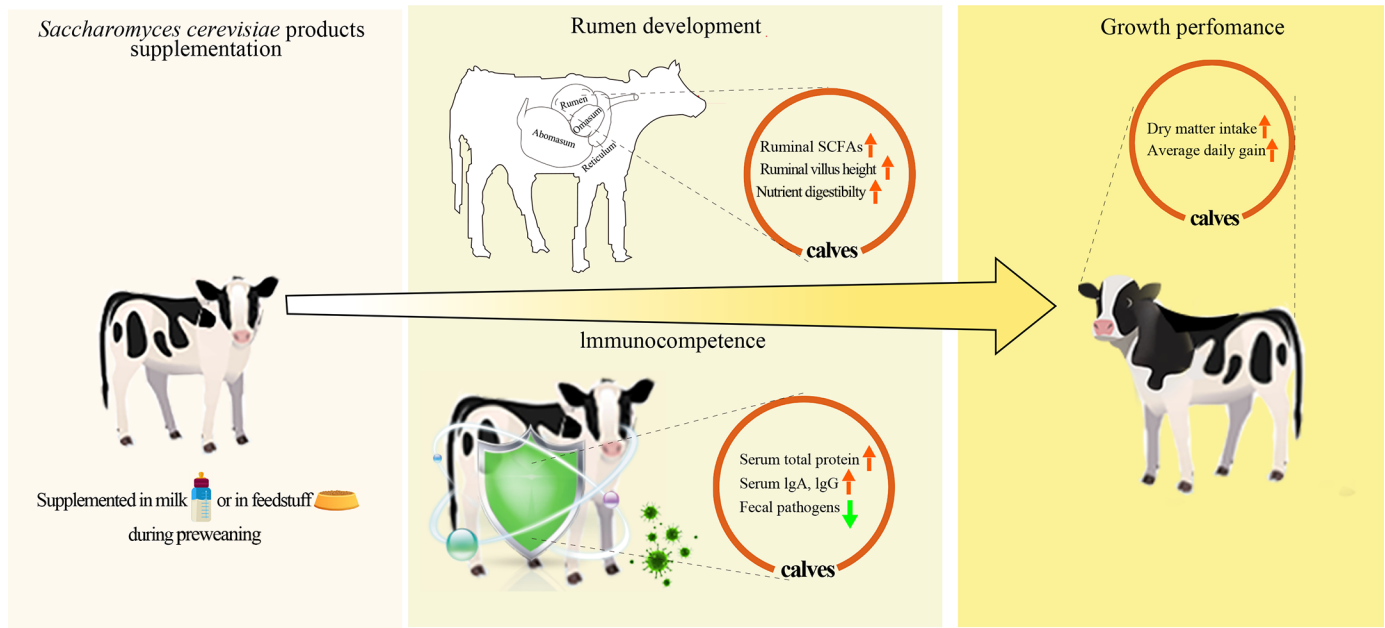


Figure 3. Summary of the effects of *Saccharomyces cerevisiae* product supplementation determined in the meta-analyses. SCFA = short-chain fatty acids.

(Lesmeister et al., 2004). However, in another study, yeast fermentation products added to the starter at a low dose (0.1% of DM) did not affect the growth performance or health condition of preweaning calves (Huuskonen and Pesonen, 2015). Hence, the dose should be considered for SCP supplementing to preweaning calves. From the dose-response curve of yeast fermentation products, we found that low or negative SMD corresponded with low supplementation doses. Considering that starter intake can vary substantially, we suggest that yeast fermentation productions should be added in an adequate amount range (around 24–25 g/d) into solid feed to achieve significant benefits. For live yeast, the goodness of fit of the models was poor ($R^2 = 0.274$ for ADG, $R^2 = 0.223$ for DMI). However, the prediction equations had some limitations, such as few observations, gaps between doses, and high variability. Hence, we expect that more data will be available in the future to obtain a better dose for using live yeast and fermentation productions in dairy calves.

CONCLUSIONS

We found that SCP supplementation could benefit calves by promoting the development of rumen papillae and microbiome and increasing circulating concentrations of immunoglobulins. Based on the results of the meta-analysis, we found that live yeast and yeast fermentation products had different effects on calves; however, there is a lack of studies to directly compare the

2 types of products on calves. The SCP supplemented to preweaning calves may have the best outcomes, especially with respect to ADG and DMI. For yeast fermentation products, 24 to 25 g/d may be a better dose added into starter feed to achieve the best ADG in preweaning calves. Future studies using high doses of SCP are needed to improve the consistency and accuracy of the dose-response curve prediction. More research is also required to investigate the long-term effects of SCP on growth performance, health, and lactation performance later in the life of dairy cows.

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CZ, JZ, and JY assessed the quality of the included trials. CZ performed the statistical analysis. JZ, GZ and JY oversaw the development of the study and resolved conflicts in the meta-analysis. CZ wrote the first draft. JZ, ZY, GZ and JY critically revised the paper. The authors read and approved the final manuscript. All data generated or analyzed during this study are included in this published article. The authors have not stated any conflicts of interest.

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